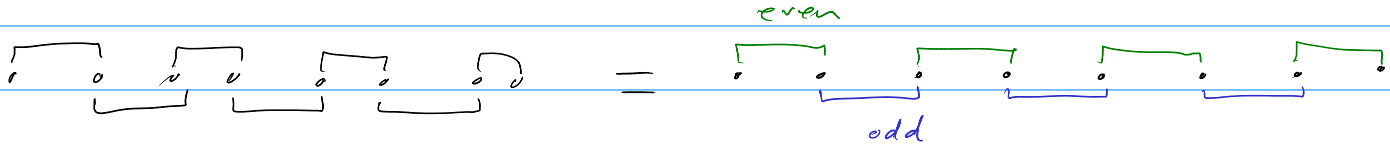
$h = -it \sum_i \gamma_i \gamma_{i+1} + g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} + \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+4}$

split hamiltonian into h_{even}, h_{odd} and rewrite majorana operators into even and odd:

$a_j := \gamma_{2n}$

$b_j := \gamma_{2n+1}$



$h = h_{even} + h_{odd}$

$= -it\left(\sum_j a_j b_j + \sum_j b_j a_{j+1}\right)$

→ Use Jordan Wigner transformation to map fermionic (majorana) operators to spin operators (Pauli matrices)

$a_j = K_j \sigma_j^x$

$b_j = K_j \sigma_j^y$

$a_j b_j = i \sigma_j^z$

$K_j := \prod_{l=1}^{j-1} \sigma_l^z$

$-it\left(\sum_j i\sigma_j^z + K_j \sigma_j^y \underline{K_{j+1}} \sigma_{j+1}^x\right)$

$= -it\left(\sum_j \left(i\sigma_j^z + K_j \sigma_j^y K_j \sigma_j^z \sigma_{j+1}^x\right)\right)$

$-it\left(\sum_j i\sigma_j^z + \cancel{K_j^2} \underbrace{\sigma_j^y \sigma_j^z}_{= i\sigma_j^x} \sigma_{j+1}^x\right)$

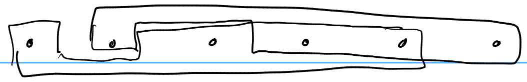
$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$

$= -it\left(\sum_j i\sigma_j^z + i\sigma_j^x \sigma_{j+1}^x\right)$

$= \sum_j -\overset{h}{\cancel{t}} \sigma_j^z - \overset{J}{\cancel{t}} \sigma_j^x \sigma_{j+1}^x = \sum_j -J \sigma_j^x \sigma_{j+1}^x - h \sigma_j^z$ – as expected.

let $t = J = h$

$$g\sum_i \gamma_i\gamma_{i+2}\gamma_{i+3}\gamma_{i+4} = g\sum_j a_j a_{j+1} b_{j+1} a_{j+2} + b_j b_{j+1} a_{j+2} b_{j+2}$$

 $=$ $a_j \quad b_j \quad a_{j+1} \quad b_{j+1} \quad a_{j+2} \quad b_{j+2}$

$$a_j a_{j+1} b_{j+1} a_{j+2} =$$

$$K_j \sigma_j^x \cdot i\sigma_j^z \cdot K_{j+2}\sigma_{j+2}^x = \cancel{K_j}\sigma_j^x\, i\sigma_{j+1}^z \cdot \cancel{K_j}\sigma_j^z\sigma_{j+1}^z\sigma_{j+2}^x$$

$$= \sigma_j^x\, i\sigma_{j+1}^z\sigma_j^z\sigma_{j+1}^z\sigma_{j+2}^x$$

$$i\sigma_j^x\sigma_j^z\underbrace{\cancel{\sigma_{j+1}^z\sigma_{j+1}^z}}_{1}\sigma_{j+2}^x$$

$$i(-i\sigma_j^y)\sigma_{j+2}^x = \sigma_j^y\sigma_{j+2}^x$$

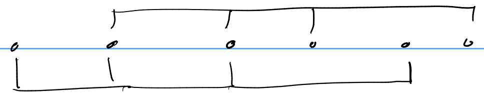
$$b_j b_{j+1} a_{j+2} b_{j+2} = \cancel{K_j}\sigma_j^y \cdot \cancel{K_j}\sigma_j^z\sigma_{j+1}^y \cdot i\sigma_{j+2}^z$$

$$\sigma_j^y\sigma_j^z\sigma_{j+1}^y\, i\sigma_{j+2}^z$$

$$\cancel{i}\, i\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z = -\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z$$

$$g\sum \gamma_i\gamma_{i+2}\gamma_{i+3}\gamma_{i+4} = g\sum \sigma_j^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z$$

$\gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+n} = \sum_j \underline{a_j b_j a_{j+1} a_{j+2}} + \underline{b_j a_{j+1} b_{j+1} b_{j+2}}$



$= a_j \quad b_j \quad a_{j+1} \quad b_{j+1} \quad a_{j+2} \quad b_{j+2}$

$a_j b_j \cdot a_{j+1} \cdot a_{j+2} = i\sigma_j^z \cdot \cancel{K_j}\, \sigma_j^z \sigma_{j+1}^x \cdot \cancel{K_j}\, \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$

$= i\sigma_j^z \cancel{\sigma_j^z \sigma_j^z}\, \sigma_{j+1}^x \sigma_{j+1}^z \sigma_{j+2}^x$ $\qquad x \to y,\ y\,z,\ z\,x \quad i\sigma_{j+1}^x$

$i\sigma_j^z (-i\sigma_{j+1}^y)\sigma_{j+2}^x = \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x$

$b_j a_{j+1} b_{j+1} b_{j+2} = \cancel{K_j}\, \sigma_j^y \cdot i\sigma_{j+1}^z \cdot \cancel{K_j}\, \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^y$

$= \sigma_j^y \sigma_j^z \left(i \underbrace{\cancel{\sigma_{j+1}^z \sigma_{j+1}^z}}_{1} \right) \sigma_{j+2}^y$

$= (i\sigma_j^x)(i)(\sigma_{j+2}^y) = -\sigma_j^x \sigma_{j+2}^y$

$g \sum \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+n} = g \sum \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y$

→ flipped sign of second term and reversed x, y, z index order
↳ makes sense.

$h = -t \sum_j \left(\sigma_j^z + \sigma_j^x \sigma_{j+1}^x \right) + g \sum \left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z \right) + \left(\sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y \right)$

$h = -t \sum_j \left(\sigma_j^z + \sigma_j^x \sigma_{j+1}^x \right) + g \sum_j \left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y \right) - \left(\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x \right)$

To study ferromagnetic phases:

$h = \sum_j J \sigma_j^x \sigma_{j+1}^x + h\sigma_j^z - g\left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y \right) - g\left(\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x \right)$

MPO: Check first term:

$$h = J\sum_j \sigma^x_j \sigma^x_{j+1} + \sigma^z_j$$

σ^x	σ^x	I ✓
I	σ^x	σ^x ✓
—x—	I	I ✓
I	I	—x—
I	σ^z	I ✓

$$\begin{array}{c|cc} 1 \quad 0 \to & \sigma^z & 0 \\ \sigma \quad 1 \to & \sigma^x & \sigma^x \\ -x-\ 2 \to & I & 0 \\ \hline & 1 & \sigma \quad -x- \\ & 0 & 1 \quad 2 \end{array} \Longrightarrow \begin{array}{c|ccc} -x- & I & 0 & 0 \\ \sigma^x & \sigma^x & 0 & 0 \\ I & \sigma^z & \sigma^x & I \\ \hline & I & \sigma^x & -x- \end{array} \qquad \begin{pmatrix} I & 0 & 0 \\ \sigma^x & 0 & 0 \\ \sigma^z & \sigma^x & I \end{pmatrix}$$

$$\begin{array}{c|ccc} 2 & & & x \\ 1 & & & x \\ & x & x & x \\ \hline & 0 & 1 & 2 \end{array} \Rightarrow$$

1

σ^z

$1 \quad \sigma^z \sigma^x$

$\sigma^z\sigma^z \quad \sigma^x$ ✓

$$\begin{pmatrix} 0 & 0 & I \\ 0 & 0 & \sigma^x \\ I & \sigma^x & \sigma^z \end{pmatrix}$$

$$g\left(\sigma^y_j \sigma^x_{j+2} - \sigma^x_j \sigma^y_{j+2}\right)$$

σ^y — o — o — σ^x or σ^x — o — o — σ^y

①

$\sigma^y I$	σ^x	I
σ^y	I	σ^x
I	σ^y	$I\sigma^x$

②

$-\sigma^x I$	σ^y	I
$-\sigma^x$	I	σ^y
I	$-\sigma^x$	$I\sigma^y$

$I \quad I \quad$ —x—

—x— $\quad I \quad I$

$$\begin{array}{c|cccccc} -x- & 1 & 0 & 0 & 0 & 0 & 0 \\ \sigma^y I & -\sigma^x & 0 & 0 & 0 & 0 & 0 \\ \sigma^x I & \sigma^y & 0 & 0 & 0 & 0 & 0 \\ \sigma^y & 0 & -1 & 0 & 0 & 0 & 0 \\ \sigma^x & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -\sigma^y & \sigma^x & 1 \\ \hline & 1 & \sigma^x & \sigma^y & \sigma^x I & \sigma^y I & -x- \end{array}$$

$$g\left(\sigma^x_j \sigma^y_{j+1} \sigma^z_{j+2} - \sigma^z_j \sigma^y_{j+1} \sigma^x_{j+2}\right)$$

I	I	—x—
I	σ^x	$\sigma^y\sigma^z$
σ^x	σ^y	σ^z
$\sigma^x\sigma^y$	σ^z	I
—x—	I	I

I	$-\sigma^z$	$\sigma^y\sigma^x$
σ^z	σ^y	σ^x
$\sigma^z\sigma^y$	σ^x	I

I

σ^x

σ^z

$\sigma^x\sigma^y$

$\sigma^z\sigma^y$

—x—

$$\begin{array}{c|cccccc} -x- & 1 & 0 & 0 & & & 0 \\ \sigma^z\sigma^y & \sigma^x & 0 & 0 & 0 & 0 & 0 \\ \sigma^x\sigma^y & -\sigma^z & 0 & 0 & 0 & 0 & 0 \\ \sigma^z & 0 & \sigma^y & 0 & 0 & 0 & 0 \\ \sigma^x & 0 & 0 & -\sigma^y & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & \sigma^z & -\sigma^x & 1 \\ \hline & 1 & \sigma^x & \sigma^z & \sigma^x\sigma^y & \sigma^z\sigma^y & -x- \end{array}$$

Combine all vectors together: